

SAT Math Mastery · Week TP1A

SAT Trick Pack 1 — Part A

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics exponent rewriting · linear constraints · tiered models · right-triangle similarity · AM–GM bounds · symmetry & Vieta · statistics effects · conditional probability · similar solids · substitution · factor roots · parameters · systems

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Rewrite radicals and mixed bases as powers of one base, then equate exponents.
- S2.** Optimise under several linear constraints: find the binding one, round in the right direction, verify the rest.
- S3.** Build a layered cost model (flat fee, tiers, surcharge, tax) and respect the order of operations.
- S4.** Use the altitude to the hypotenuse: similar triangles, geometric means, area ratios.
- S5.** Bound a product with a fixed sum of squares ($a^2 + b^2 \geq 2ab$) and find where equality holds.
- S6.** Read the axis of symmetry from equal outputs; combine with Vieta and sign facts.
- S7.** Decide how a replacement changes the mean and the spread without recomputing.
- S8.** Condition on a subgroup (it is the new denominator) and recover a missing count.
- S9.** Scale area by k^2 and volume by k^3 ; invert the factor from an area ratio.
- S10.** Substitute $u = a^x$ to turn an exponential equation into a quadratic; reject impossible roots.
- S11.** Plug in a zero of a factor instead of expanding.
- S12.** Parameters in linear equations, absolute-value graphs and parametrised solution sets.

2 · REQUIRED FACTS

- $\sqrt[n]{a^m} = a^{m/n}$, $a^m a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$; if $a > 0$, $a \neq 1$ and $a^p = a^q$ then $p = q$.
- Exponentials a^x with $a > 0$ are always positive: $a^x = \text{negative}$ has no real solution.
- Altitude to the hypotenuse splits a right triangle into two triangles similar to it and to each other.
- Replacing values keeps the mean if the total is unchanged; standard deviation falls when extreme values move toward the centre.
- Conditioning on a group makes that group's total the denominator.
- Similar solids with linear factor k : lengths $\times k$, areas $\times k^2$, volumes $\times k^3$.
- If $f(p) = f(q)$ for a quadratic, the axis is $x = \frac{p+q}{2}$; zeros sum to $-\frac{b}{a}$ and multiply to $\frac{c}{a}$.
- A linear equation whose x -terms cancel has either no solution or infinitely many, decided by the leftover constants.

3 · REQUIRED FORMULAS

$$a^{m/n} = \sqrt[n]{a^m}, \quad CD^2 = AD \cdot DB, \quad AC^2 = AD \cdot AB, \quad BC^2 = DB \cdot AB, \quad \frac{[ACD]}{[CDB]} = \frac{AD}{DB}$$

$$a^2 + b^2 \geq 2ab, \quad x_{\text{axis}} = \frac{p+q}{2} = -\frac{b}{2a}, \quad r+s = -\frac{b}{a}, \quad rs = \frac{c}{a}, \quad \frac{V_2}{V_1} = \left(\sqrt{\frac{S_2}{S_1}} \right)^3$$

4 · MINI THEORY SHEET

Common base first. Convert every radical and every base to one prime base ($4 = 2^2$, $8 = 2^3$, $\sqrt[3]{2^x} = 2^{x/3}$), simplify each side to a single power, and only then equate exponents. Two equations become a linear system in the exponents.

Constraint problems. Turn each sentence into an inequality. Test the maximum (or minimum) against *every* constraint; the one that fails first is binding. Whole-number answers: round down for a maximum, up for a minimum, then re-check.

Layered models. Write the pipeline in the order described (fee $\rightarrow$ tiers $\rightarrow$ surcharge on the stated part $\rightarrow$ tax on the stated total). Test one input by hand and compare with the formulas.

Parameters. Expand and collect. Same x -coefficient on both sides leaves a statement about constants: true means infinitely many solutions, false means none.

Substitution. If a^x appears as a^{2x} and a^x (or $a^{x+c} = a^c \cdot a^x$), set $u = a^x$ with $u > 0$.

Speed shortcuts.

Situation	Shortcut
mixed radicals and bases	one common base, then equate exponents; solve the linear system
ratio-plus-budget constraints	set the ratio constraint to equality, test each other limit
fee, tiers, surcharge, tax	one test input by hand; compare with each option
altitude to the hypotenuse	area ratio = segment ratio; $CD^2 = AD \cdot DB$
product with fixed $a^2 + b^2$	maximum at $a = b$ (rescale if a coefficient sits in front)
equal outputs of a quadratic	axis at the midpoint of the two inputs
swap values, same total	mean unchanged; spread falls if extremes move inward
“of those who ...”	the subgroup total is the denominator
areas $\rightarrow$ volumes	$k = \sqrt{\text{area ratio}}$, then k^3
A^{2x} and A^x together	$u = A^x$, factor, discard $u \leq 0$
$p(x) = (x - r)q(x)$ with unknown coefficient	substitute $x = r$
$ax + b = cx + d$ with parameter	match coefficients: same line $\Rightarrow$ infinitely many
$ x - h + v$ against a line	one solution exactly when the line hits the vertex
equations that are multiples	substitute the parametrised point into one equation: do the t -terms cancel?

5 · GUIDED EXAMPLES

Example 1 — Common base, then a linear system

If $2^x \cdot 8^y = 2^9$ and $\frac{4^x}{2^y} = 16$, find x and y .

$$2^x \cdot 2^{3y} = 2^9 \Rightarrow x + 3y = 9. \quad \frac{2^{2x}}{2^y} = 2^4 \Rightarrow 2x - y = 4.$$

From the second, $y = 2x - 4$; then $x + 6x - 12 = 9 \Rightarrow x = 3$, $y = 2$.

Example 2 — Similar solids: area $\rightarrow$ factor $\rightarrow$ volume

Two similar solids have surface areas 36 and 81. The smaller has volume 40. Find the volume of the larger.

$$k = \sqrt{\frac{81}{36}} = \frac{3}{2}. \quad \text{Volume scales by } k^3 = \frac{27}{8}: \quad 40 \cdot \frac{27}{8} = 135.$$

Example 3 — Equal outputs fix the axis

$g(x) = x^2 + bx + c$, with $g(2) = g(12)$ and $g(0) = 11$. Find the minimum value of g .

$$\text{Axis: } x = \frac{2+12}{2} = 7 = -\frac{b}{2} \Rightarrow b = -14. \quad c = g(0) = 11.$$

$$\text{Minimum} = g(7) = 49 - 98 + 11 = -38.$$

6 · PRACTICE SET — 15 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · CALCULATOR

Real numbers x and y satisfy

$$\sqrt[3]{2^x} \cdot 4^y = 256 \quad \text{and} \quad \frac{16^x}{\sqrt[3]{4^y}} = 2^{22}.$$

What is the value of xy ?

- Write everything as powers of 2: $\sqrt[3]{2^x} = 2^{x/3}$, $4^y = 2^{2y}$, $256 = 2^8$. First equation: $2^{x/3+2y} = 2^8$, so $\frac{x}{3} + 2y = 8$, i.e. $x + 6y = 24$.
- Second equation: $16^x = 2^{4x}$ and $\sqrt[3]{4^y} = 2^{2y/3}$, so $2^{4x-2y/3} = 2^{22}$, giving $4x - \frac{2y}{3} = 22$, i.e. $6x - y = 33$.
- From $6x - y = 33$: $y = 6x - 33$. Substitute: $x + 6(6x - 33) = 24 \Rightarrow 37x = 222 \Rightarrow x = 6$, so $y = 3$.
- Check: $\frac{6}{3} + 2(3) = 8$ and $4(6) - \frac{2(3)}{3} = 22$.
- $xy = 6 \cdot 3 = 18$.

Answer: 18

2.

SAT CHALLENGE · CALCULATOR

A science department orders microscopes at \$240 each and slide kits at \$15 each. The order must include at least 6 slide kits for every microscope, the total cost may not exceed \$2,500, and the storage cabinet holds at most 46 items (microscopes and kits combined). What is the greatest number of microscopes the department can order?

- Let m = microscopes and k = kits. Constraints: $k \geq 6m$, $240m + 15k \leq 2500$, $m + k \leq 46$.
- Taking the fewest kits allowed, $k = 6m$, makes both the cost and the item count as small as possible for a given m , so test $k = 6m$.
- Budget: $240m + 90m = 330m \leq 2500 \Rightarrow m \leq 7.57$, so $m \leq 7$.
- Cabinet: $m + 6m = 7m \leq 46 \Rightarrow m \leq 6.57$, so $m \leq 6$. This is the tighter limit.
- Check $m = 6$, $k = 36$: cost $1440 + 540 = 1980 \leq 2500$, items $42 \leq 46$. For $m = 7$ the cabinet needs at least 49 items, which fails.
- The greatest number is 6.

Answer: 6

3.

SAT CHALLENGE · CALCULATOR

A utility charges a \$15 service fee each month. Usage is billed at \$2.40 per unit for the first 20 units and \$3.60 for each unit beyond 20. A 10% fuel surcharge is added to the usage charge only (not to the service fee). Then 5% tax is applied to the sum of the service fee and the surcharged usage charge. For $u > 20$ units, which function gives the total monthly bill $B(u)$, in dollars?

(A) $B(u) = 3.78u - 9.45$

(B) $B(u) = 4.158u - 11.97$ ✓

(C) $B(u) = 4.158u - 10.395$

(D) $B(u) = 4.158u - 12.72$

- Usage charge for $u > 20$: $2.40(20) + 3.60(u - 20) = 48 + 3.6u - 72 = 3.6u - 24$.
- Surcharge on usage only: $1.10(3.6u - 24) = 3.96u - 26.4$.
- Add the service fee: $15 + 3.96u - 26.4 = 3.96u - 11.4$.
- Tax on that sum: $1.05(3.96u - 11.4) = 4.158u - 11.97$.
- Check with $u = 30$: usage $48 + 36 = 84$; surcharged 92.40 ; plus fee 107.40 ; with tax 112.77 . And $4.158(30) - 11.97 = 112.77$.

Answer: (B)

4.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The area of triangle ACD is 48 and the area of triangle CDB is 108. What is the length of $\overline{AB}$?

- Triangles ACD and CDB share the height CD , so their areas are in the ratio of their bases: $\frac{AD}{DB} = \frac{48}{108} = \frac{4}{9}$.
- Let $AD = 4k$ and $DB = 9k$. In the right triangle ABC the altitude satisfies $CD^2 = AD \cdot DB = 36k^2$, so $CD = 6k$.
- Area of ACD : $\frac{1}{2}(4k)(6k) = 12k^2 = 48 \Rightarrow k^2 = 4 \Rightarrow k = 2$.
- $AB = AD + DB = 13k = 26$. (Check: $[CDB] = \frac{1}{2}(18)(12) = 108$.)

Answer: 26

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers p and q with $p^2 + 4q^2 = 72$, the product $(2x + p)(x + q)$ is equivalent to $2x^2 + bx + c$, where b and c are constants. What is the greatest possible value of c ?

1. Expand: $(2x + p)(x + q) = 2x^2 + (2q + p)x + pq$, so $c = pq$.
2. Write $p^2 + 4q^2 = p^2 + (2q)^2$. Since $(p - 2q)^2 \geq 0$, we have $p^2 + (2q)^2 \geq 2p(2q) = 4pq$.
3. Therefore $4pq \leq 72$, so $pq \leq 18$.
4. Equality needs $p = 2q$: $p = 6$, $q = 3$ gives $36 + 36 = 72$ and $pq = 18$. The greatest value of c is 18.

Answer: 18

6.

SAT CHALLENGE • NO CALCULATOR NEEDED

The quadratic function $f(x) = ax^2 + bx + c$, where a , b and c are constants, has two distinct real zeros. It is known that $f(2) = f(8)$, $f(0) < 0$ and $f(5) > 0$. Consider the following statements.

- I. $b > 0$
- II. The product of the two zeros is greater than 25.
- III. Both zeros are positive.

Which of the statements must be true?

- (A) I only
- (B) II and III only
- (C) I and III only ✓
- (D) I, II, and III

1. $f(2) = f(8)$ puts the axis at $x = 5$, so $-\frac{b}{2a} = 5$ and $b = -10a$. The zeros sum to $-\frac{b}{a} = 10$.
2. $f(5) > 0$ says the vertex is above the x -axis. A parabola with two zeros and a vertex above the axis opens downward, so $a < 0$.
3. I: $b = -10a > 0$. True.
4. II: zeros r, s with $r + s = 10$ and $r \neq s$ give $rs < \left(\frac{r+s}{2}\right)^2 = 25$. False (the product is always less than 25).
5. III: $f(0) = c < 0$ and $a < 0$, so $rs = \frac{c}{a} > 0$; with $r + s = 10 > 0$ both zeros are positive. True.
6. Only I and III must be true.

Answer: (C)

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has eight values: 12, 15, 18, 20, 20, 22, 25, 28. The values 12 and 28 are removed and the values 19 and 21 are added in their place. Which of the following describes the mean and the standard deviation (SD) of the new data set compared with the original data set?

(A) Mean smaller, SD smaller

(B) Mean same, SD larger

(C) Mean same, SD smaller ✓

(D) Mean same, SD same

1. Total removed: $12 + 28 = 40$. Total added: $19 + 21 = 40$. The sum is still 160, so the mean is still 20 (unchanged).
2. The two extreme values, 8 below and 8 above the mean, were replaced by values only 1 from the mean. With the mean fixed, the data are more tightly clustered, so the standard deviation is smaller. (Check: original SD ≈ 4.27 , new SD ≈ 2.87 .)
3. Mean unchanged, standard deviation smaller.

Answer: (C)

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

A conference has 300 participants. Each participant attended either the Technical track or the Business track, and either in person or online. Some of the data are shown.

	Technical	Business	Total
In person	70		
Online	50		
Total	120		300

Of the participants in the Business track, 35% attended online. How many of the 300 participants attended in person?

(A) 187 ✓

(B) 63

(C) 113

(D) 145

1. Business track total: $300 - 120 = 180$.

2. Business participants online: $0.35 \times 180 = 63$, so Business participants in person: $180 - 63 = 117$.

3. In person overall: $70 + 117 = 187$.

4. (The percentage applies to the 180 Business participants, not to all 300.)

Answer: (A)

9.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two similar rectangular crates have surface areas of 144 square feet and 400 square feet. The smaller crate has a volume of 108 cubic feet. What is the volume, in cubic feet, of the larger crate?

(A) 1500

(B) 180

(C) 300

(D) 500 ✓

1. Areas scale by k^2 : $k^2 = \frac{400}{144} = \frac{25}{9}$, so the linear scale factor is $k = \frac{5}{3}$.

2. Volumes scale by $k^3 = \frac{125}{27}$.

3. $108 \cdot \frac{125}{27} = 4 \cdot 125 = 500$.

4. (Check with dimensions $3 \times 6 \times 6$: surface area 144, volume 108; scaled by $\frac{5}{3}$: $5 \times 10 \times 10$, surface area 400, volume 500.)

Answer: (D)

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number x , $4^x - 2^{x+2} - 32 = 0$. What is the value of 2^{x+1} ?

(A) -8 (B) 8 **(C) 16 ✓**(D) 32

1. Write $4^x = (2^x)^2$ and $2^{x+2} = 4 \cdot 2^x$. Let $u = 2^x$ (so $u > 0$): $u^2 - 4u - 32 = 0$.
2. Factor: $(u - 8)(u + 4) = 0$, so $u = 8$ or $u = -4$.
3. 2^x is always positive, so $u = -4$ is rejected. $u = 8$, i.e. $x = 3$.
4. $2^{x+1} = 2u = 16$.

Answer: (C)**11.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For constants p , q and c , the equation

$$(2x - 3)(x^2 + px + q) = 2x^3 - 5x^2 + cx + 9$$

is true for all real values of x . What is the value of c ?

1. The left side is 0 when $x = \frac{3}{2}$, so the right side must be 0 there too.
2. $2\left(\frac{27}{8}\right) - 5\left(\frac{9}{4}\right) + c\left(\frac{3}{2}\right) + 9 = 0$.
3. $\frac{27}{4} - \frac{45}{4} + 9 + \frac{3}{2}c = 0 \Rightarrow \frac{9}{2} + \frac{3}{2}c = 0 \Rightarrow c = -3$.
4. (Check: $p = -1$, $q = -3$ gives $(2x - 3)(x^2 - x - 3) = 2x^3 - 5x^2 - 3x + 9$.)

Answer: -3

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two similar solid blocks have surface areas of 208 square centimeters and 325 square centimeters. The larger block has a volume of 375 cubic centimeters. What is the volume, in cubic centimeters, of the smaller block?

1. Going from the larger to the smaller block, areas scale by $\frac{208}{325} = \frac{16}{25}$.

2. So the linear factor is $\sqrt{\frac{16}{25}} = \frac{4}{5}$.

3. Volume scales by $\left(\frac{4}{5}\right)^3 = \frac{64}{125}$.

4. $375 \cdot \frac{64}{125} = 3 \cdot 64 = 192$.

Answer: 192

13.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$k(x + 3) - 5x = 3x + 2m$$

In the equation above, k and m are constants. If the equation has infinitely many solutions, what is the value of $m - k$?

(A) 20

(B) 4 ✓

(C) 8

(D) 12

1. Expand and collect: $(k - 5)x + 3k = 3x + 2m$.

2. Infinitely many solutions require the x -terms to match and the constants to match: $k - 5 = 3 \Rightarrow k = 8$, and $3k = 2m \Rightarrow 24 = 2m \Rightarrow m = 12$.

3. $m - k = 12 - 8 = 4$. (The options 8, 12 and 20 are k , m and $k + m$.)

Answer: (B)

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

The equation $|x + 6| - 7 = \frac{x}{3} + c$, where c is a constant, has exactly one solution x . What is the value of $2c + x$?

(A) -11 (B) -6 (C) -5 (D) -16 ✓

1. The graph of $y = |x + 6| - 7$ is a V with vertex $(-6, -7)$ and arms of slope ± 1 . The line $y = \frac{x}{3} + c$ is less steep than either arm.
2. Such a line meets the V twice if it passes above the vertex, never if it passes below, and exactly once only if it passes through the vertex.
3. Through $(-6, -7)$: $-7 = \frac{-6}{3} + c \Rightarrow c = -5$, and the solution is $x = -6$.
4. $2c + x = -10 - 6 = -16$.
5. (Algebra check: for $x \geq -6$, $x = \frac{3(c+1)}{2}$ is valid when $c \geq -5$; for $x < -6$, $x = -\frac{3(c+13)}{4}$ is valid when $c > -5$. Exactly one solution occurs only at $c = -5$.)

Answer: (D)

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$3x + 2y = 12$$

$$9x + 6y = 36$$

For the system above, which of the following ordered pairs (x, y) , where t is any real number, is a solution to the system?

(A) $(2t + 4, -3t)$ ✓(B) $(2t + 4, 3t)$ (C) $(3t + 4, -2t)$ (D) $(4 - 2t, -3t)$

1. The second equation is 3 times the first, so the two equations describe the same line. A pair is a solution exactly when it satisfies $3x + 2y = 12$ for every t .
2. (A): $3(2t + 4) + 2(-3t) = 6t + 12 - 6t = 12$. The t -terms cancel. ✓
3. (B): $3(2t + 4) + 2(3t) = 12t + 12$. Not constant. (C): $3(3t + 4) + 2(-2t) = 5t + 12$. Not constant. (D): $3(4 - 2t) + 2(-3t) = 12 - 12t$. Not constant.
4. Only $(2t + 4, -3t)$ works.

Answer: (A)